

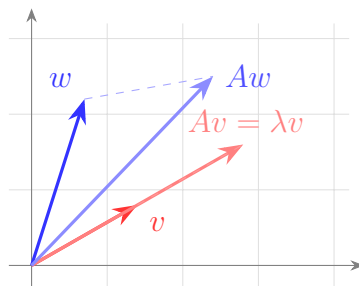
Exercise Sheet 11 – Solutions

Linear Algebra

What do the terms mean?

- **Eigenvalue λ and eigenvector v :** a number λ and a vector $v \neq 0$ with $Av = \lambda v$. That means: A only *stretches* the vector v (by the factor λ), without changing its direction.
- **Characteristic polynomial $\chi_A(t) = \det(A - tI)$:** the roots of this polynomial are exactly the eigenvalues. (Reason: $Av = \lambda v$ means $(A - \lambda I)v = 0$ with $v \neq 0$, so $A - \lambda I$ is *not* invertible, hence $\det(A - \lambda I) = 0$.)
- **Eigenspace $E_\lambda = \ker(A - \lambda I)$:** all eigenvectors for λ (plus the zero vector). You find it by solving the system $(A - \lambda I)v = 0$.
- **Algebraic multiplicity $\text{AM}(\lambda)$:** how often λ appears as a root of the char. polynomial (the exponent of $(t - \lambda)$).
- **Geometric multiplicity $\text{GM}(\lambda)$:** $\dim E_\lambda =$ number of basis vectors of the eigenspace = number of free variables when solving $(A - \lambda I)v = 0$.
- **It always holds that $1 \leq \text{GM}(\lambda) \leq \text{AM}(\lambda)$.** If all $\text{GM} = \text{AM}$, then A is *diagonalizable*.

[Graphical explanation – not part of the solution]



*The basic idea: multiplying a vector by A usually rotates and stretches it (blue: $w \rightarrow Aw$ points elsewhere). An **eigenvector** (red) is a special direction that A only stretches – Av lies on the same line as v . The stretch factor is the eigenvalue λ .*

Exercise 43

Problem. For $A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}$ determine the characteristic polynomial, the eigenvalues with algebraic and geometric multiplicity, and a basis of each eigenspace – in **two ways**.

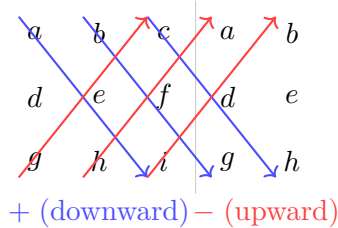
What needs to be done?

- **Method 1:** multiply out $\chi_A(t) = \det(A - tI)$ with the **Sarrus rule**, then factor with the **theorem on integer roots** (candidates = divisors of the constant term).
- **Method 2:** use **row/column operations** to factor out a term $(\lambda - t)$ early – this saves the multiplying-out.
- Then, per eigenvalue λ , solve the system $(A - \lambda I)v = 0 \Rightarrow$ basis of the eigenspace and GM.

Matrix A

Method 1: Sarrus + integer roots.

[Graphical explanation – not part of the solution]



Sarrus rule: write the first two columns again on the right. The three diagonals going down (blue) count +, the three going up (red) count -. Formula: $\det = aei + bfg + cdh - ceg - afh - bdi$.

Step 1: Write down $A - tI$. tI is the diagonal matrix with t all along the diagonal. So subtracting it changes *only the diagonal* of A :

$$A - tI = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}.$$

Step 2: Apply Sarrus. With the formula above (the three + diagonals, then the three - diagonals):

$$\begin{aligned} \det(A - tI) &= \underbrace{(4-t)(5-t)(2-t)}_{aei} + \underbrace{1 \cdot 2 \cdot (-1)}_{bfg} + \underbrace{1 \cdot 2 \cdot (-1)}_{cdh} \\ &\quad - \underbrace{1 \cdot (5-t) \cdot (-1)}_{ceg} - \underbrace{(4-t) \cdot 2 \cdot (-1)}_{afh} - \underbrace{1 \cdot 2 \cdot (2-t)}_{bdi} \\ &= (4-t)(5-t)(2-t) - 2 - 2 + (5-t) + 2(4-t) - 2(2-t). \end{aligned}$$

The bracket term multiplied out: $(4-t)(5-t)(2-t) = -t^3 + 11t^2 - 38t + 40$. The remaining terms: $-2 - 2 + 5 - t + 8 - 2t - 4 + 2t = 5 - t$. Together:

$$\chi_A(t) = -t^3 + 11t^2 - 38t + 40 + 5 - t = -t^3 + 11t^2 - 39t + 45.$$

[Explanation added] Quick check: constant term $\chi_A(0) = 45 = \det A \checkmark$, and the t^2 -coefficient $11 = \text{trace}(A) = 4 + 5 + 2 \checkmark$.

Step 3: Factor with integer roots. Factor out -1 : $\chi_A(t) = -(t^3 - 11t^2 + 39t - 45)$. Integer roots divide the constant term 45 (candidates $\pm 1, 3, 5, 9, 15, 45$). Substituting:

$$t = 3 : 27 - 99 + 117 - 45 = 0 \checkmark, \quad t = 5 : 125 - 275 + 195 - 45 = 0 \checkmark.$$

The third root from the trace: $3 + 5 + \lambda_3 = 11 \Rightarrow \lambda_3 = 3$. So

$$\chi_A(t) = -(t-3)^2(t-5) = (3-t)^2(5-t).$$

Eigenvalues: $\lambda = 3$ (AM 2), $\lambda = 5$ (AM 1).

Method 2: Factor out early (column trick).

[Graphical explanation – not part of the solution]

$$\begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2 + R_3} \begin{pmatrix} 5-t & 5-t & 5-t \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}$$

Observation: in A every column sums to 5 (e.g. $4 + 2 - 1 = 5$). So in $A - tI$ every column sums to $5 - t$. If you add all rows together ($R_1 \rightarrow R_1 + R_2 + R_3$), the top row becomes all $(5 - t)$ – which you can pull out as a factor.

Step 1: $R_1 \rightarrow R_1 + R_2 + R_3$ (does not change the determinant), then factor out $(5 - t)$:

$$\det(A - tI) = \det \begin{pmatrix} 5-t & 5-t & 5-t \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix} = (5-t) \det \begin{pmatrix} 1 & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}.$$

Step 2: columns $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ (makes the first row 1, 0, 0):

$$= (5-t) \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3-t & 0 \\ -1 & 0 & 3-t \end{pmatrix} = (5-t)(3-t)(3-t) = (5-t)(3-t)^2.$$

[Explanation added] The last matrix is (lower) triangular $\Rightarrow$ determinant = product of the diagonal $1 \cdot (3-t) \cdot (3-t)$.

Same result as Method 1: $\chi_A(t) = (5-t)(3-t)^2$, eigenvalues $\lambda = 5$ (AM 1), $\lambda = 3$ (AM 2).

Eigenspaces.

$\lambda = 5$: solve $(A - 5I)v = 0$. First $A - 5I$ (only the diagonal decreases by 5):

$$A - 5I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 + 2R_1} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Free variable z : from $2y + 4z = 0 \Rightarrow y = -2z$; from $-x + y + z = 0 \Rightarrow x = y + z = -z$.
With $z = 1$:

$$E_5 = \text{span}\{(-1, -2, 1)\}, \quad \text{GM}(5) = 1.$$

$\lambda = 3$: First $A - 3I$:

$$A - 3I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Only one equation $x + y + z = 0$, free variables y, z . With $(y, z) = (1, 0)$ and $(0, 1)$:

$$E_3 = \text{span}\{(-1, 1, 0), (-1, 0, 1)\}, \quad \text{GM}(3) = 2.$$

$$\chi_A(t) = (5 - t)(3 - t)^2.$$

$\lambda = 5$: AM 1, GM 1, basis $\{(-1, -2, 1)\}$. $\lambda = 3$: AM 2, GM 2, basis $\{(-1, 1, 0), (-1, 0, 1)\}$.

All GM = AM $\Rightarrow A$ is diagonalizable.

Matrix B

Method 1: Sarrus. First $B - tI$ (again only the diagonal):

$$B - tI = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 5-t & 2 & 3 \\ 2 & 4-t & 2 \\ -3 & -2 & -1-t \end{pmatrix}.$$

With this the Sarrus rule gives (same scheme as above):

$$\chi_B(t) = (5 - t)(4 - t)(-1 - t) + 36 - 9t = -t^3 + 8t^2 - 20t + 16.$$

[Explanation added] Check: $\chi_B(0) = 16 = \det B$ ✓, t^2 -coefficient $8 = \text{trace}(B) = 5 + 4 - 1$ ✓.

Factor: $\chi_B(t) = -(t^3 - 8t^2 + 20t - 16)$. Test divisors of 16: $t = 2$: $8 - 32 + 40 - 16 = 0$ ✓. Polynomial division by $(t - 2)$:

$$t^3 - 8t^2 + 20t - 16 = (t - 2)(t^2 - 6t + 8) = (t - 2)(t - 2)(t - 4).$$

$$\chi_B(t) = -(t - 2)^2(t - 4) = (2 - t)^2(4 - t).$$

Method 2: column trick. In B too every column sums to 4 ($5 + 2 - 3 = 4$, $2 + 4 - 2 = 4$, $3 + 2 - 1 = 4$). With $R_1 \rightarrow R_1 + R_2 + R_3$ and then $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$:

$$\det(B - tI) = (4 - t) \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2-t & 0 \\ -3 & 1 & 2-t \end{pmatrix} = (4 - t)(2 - t)^2.$$

Same result. Eigenvalues: $\lambda = 2$ (AM 2), $\lambda = 4$ (AM 1).

Eigenspaces. $\lambda = 4$: First $B - 4I$:

$$B - 4I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 3R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 0 & 0 \end{pmatrix}.$$

 $-4y - 4z = 0 \Rightarrow y = -z$; $x + 2y + 3z = 0 \Rightarrow x = -2y - 3z = -z$. With $z = 1$:

$$E_4 = \text{span}\{(-1, -1, 1)\}, \quad \text{GM}(4) = 1.$$

 $\lambda = 2$: First $B - 2I$:

$$B - 2I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \begin{pmatrix} 3 & 2 & 3 \\ 1 & 1 & 1 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 3R'_2]{R_1 \rightarrow R_1 - 3R'_2} \begin{pmatrix} 0 & -1 & 0 \\ \boxed{1} & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

From the (-1) row: $-y = 0 \Rightarrow y = 0$; from $x + y + z = 0 \Rightarrow x = -z$. Only free variable z . With $z = 1$:

$$E_2 = \text{span}\{(-1, 0, 1)\}, \quad \text{GM}(2) = 1.$$

$$\chi_B(t) = (2 - t)^2(4 - t).$$

$\lambda = 2$: AM 2, GM 1, basis $\{(-1, 0, 1)\}$. $\lambda = 4$: AM 1, GM 1, basis $\{(-1, -1, 1)\}$.

Here $\text{GM}(2) = 1 < \text{AM}(2) = 2 \Rightarrow B$ is *not* diagonalizable.

Exercise 44**Problem.** Determine the eigenvalues (with AM, GM) and bases of the eigenspaces of

$$A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} \text{ and } B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix}.$$

What needs to be done?

- **No characteristic polynomial!** Instead use properties: $\lambda = 0$ exactly when A is not invertible; sum of eigenvalues = trace; product = det; $\text{GM} \leq \text{AM}$; read off obvious eigenvectors.

Matrix A

[Graphical explanation – not part of the solution]

$$A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} \quad \begin{array}{l} \text{row 1} \\ Ae_2 = (0, 6, 0) = 6e_2 \Rightarrow \lambda = 6 \\ \text{row 3} \Rightarrow \det = 0 \Rightarrow \lambda = 0 \end{array}$$

Two things are immediate: (1) row 1 = row 3, so A is singular $\Rightarrow 0$ is an eigenvalue. (2) The middle column is $(0, 6, 0)$, i.e. A sends $e_2 = (0, 1, 0)$ to $6e_2 \Rightarrow 6$ is an eigenvalue with eigenvector e_2 .

Step 1: $\lambda = 0$. Row 1 = row 3 $\Rightarrow \det A = 0 \Rightarrow \lambda = 0$ is an eigenvalue. Eigenspace = $\ker A$:

$$Av = 0 : \quad 6y = 0 \Rightarrow y = 0, \quad x + 5z = 0 \Rightarrow x = -5z.$$

$$E_0 = \text{span}\{(-5, 0, 1)\}, \quad \text{GM}(0) = 1.$$

Step 2: $\lambda = 6$. Eigenspace = $\ker(A - 6I)$. First

$$A - 6I = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} - \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} -5 & 0 & 5 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Both nontrivial rows say $x = z$ (y free). So

$$E_6 = \text{span}\{(0, 1, 0), (1, 0, 1)\}, \quad \text{GM}(6) = 2.$$

Step 3: Add up the multiplicities. We have $\text{GM}(6) = 2$, so $\text{AM}(6) \geq 2$. Plus $\text{AM}(0) \geq 1$. Since a 3×3 matrix has exactly 3 eigenvalues (with multiplicity), only $\text{AM}(0) = 1$, $\text{AM}(6) = 2$ remain. [Explanation added] Check with the trace: $0 \cdot 1 + 6 \cdot 2 = 12 = \text{trace}(A) = 1 + 6 + 5 \checkmark$.

$$\lambda = 0: \text{AM } 1, \text{GM } 1, \text{basis } \{(-5, 0, 1)\}. \quad \lambda = 6: \text{AM } 2, \text{GM } 2, \text{basis } \{(0, 1, 0), (1, 0, 1)\}.$$

All $\text{GM} = \text{AM} \Rightarrow A$ is diagonalizable.

Matrix B

[Graphical explanation – not part of the solution]

$$B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix} \quad \left\{ \begin{array}{l} \text{every row} = c_i \cdot (1, 1, 1, 1) \\ c = (5, 7, 11, -23) \\ \Rightarrow \text{rank } 1 \end{array} \right.$$

B has identical columns everywhere – every row is a multiple of $(1, 1, 1, 1)$. Such a matrix has **rank** 1: all rows lie “on one line”. Hence the kernel is huge (dimension $4 - 1 = 3$) and 0 is an eigenvalue.

Step 1: $\lambda = 0$ with GM = 3. All columns of B are equal $\Rightarrow \text{rank}(B) = 1 \Rightarrow \dim \ker(B) = 4 - 1 = 3$. So 0 is an eigenvalue with $\text{GM}(0) = 3$. The kernel: $Bv = 0$ means (each row) $5(a + b + c + d) = 0$, i.e. $a + b + c + d = 0$:

$$E_0 = \text{span}\{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}, \quad \text{GM}(0) = 3.$$

Step 2: There is no further eigenvalue. The sum of all eigenvalues = $\text{trace}(B) = 5 + 7 + 11 - 23 = 0$. So far we have $\lambda = 0$ with $\text{AM}(0) \geq \text{GM}(0) = 3$. If the fourth eigenvalue μ were $\neq 0$, the sum would be $\mu \neq 0$ – contradiction. So $\mu = 0$ and

$$\text{AM}(0) = 4.$$

[Explanation added] Put differently: a rank-1 matrix $cr^\top$ has only the eigenvalue $r^\top c = \text{trace}$ (once) and otherwise 0. Here $\text{trace} = 0$, so *all* eigenvalues are 0 (the matrix is nilpotent, $B^2 = 0$).

$$\lambda = 0: \text{AM } 4, \text{ GM } 3, \text{ basis } \{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}.$$

$$\text{GM}(0) = 3 < \text{AM}(0) = 4 \Rightarrow B \text{ is not diagonalizable.}$$

Exercise 45

Problem. $A, B \in \mathbb{F}^{n \times n}$. Prove the following claims.

What needs to be done?

- In each claim we start with “ λ is an eigenvalue of A ” = “there is $v \neq 0$ with $Av = \lambda v$ ” and check that the same v is also an eigenvector of the new matrix.

(a) If λ is an eigenvalue of A and $c \in \mathbb{F}$, then $\lambda + c$ is an eigenvalue of $A + cI$.

Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$(A + cI)v = Av + cv = \lambda v + cv = (\lambda + c)v.$$

Since $v \neq 0$, $\lambda + c$ is an eigenvalue of $A + cI$ (with the same eigenvector v). □

(b) If λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2 .

Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2v. \quad \square$$

(c) If A is invertible and λ is an eigenvalue of A , then λ^{-1} is an eigenvalue of A^{-1} .

Proof. Since A is invertible, 0 is not an eigenvalue (otherwise there would be $v \neq 0$ with $Av = 0$, so A would not be injective). Hence $\lambda \neq 0$. From $Av = \lambda v$, applying A^{-1} :

$$v = A^{-1}(\lambda v) = \lambda A^{-1}v \implies A^{-1}v = \frac{1}{\lambda}v.$$

Since $v \neq 0$, λ^{-1} is an eigenvalue of A^{-1} . $\square$

(d) If v is an eigenvector of both A and B , then v is also an eigenvector of AB .

Proof. Let $v \neq 0$ with $Av = \lambda v$ and $Bv = \mu v$. Then

$$(AB)v = A(Bv) = A(\mu v) = \mu(Av) = \mu(\lambda v) = (\lambda\mu)v.$$

Since $v \neq 0$, v is an eigenvector of AB for the eigenvalue $\lambda\mu$. $\square$

Overview: Results 43 & 44

Matrix	Eigenvalue (λ)	AM, GM	Basis of the eigenspace
43 A	3	2, 2	$(-1, 1, 0), (-1, 0, 1)$
	5	1, 1	$(-1, -2, 1)$
43 B	2	2, 1	$(-1, 0, 1)$
	4	1, 1	$(-1, -1, 1)$
44 A	0	1, 1	$(-5, 0, 1)$
	6	2, 2	$(0, 1, 0), (1, 0, 1)$
44 B	0	4, 3	$(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)$

Diagonalizable are 43 A and 44 A (all GM = AM). *Not diagonalizable* are 43 B and 44 B (one GM < AM each).